

Git Essentials: Command Reference Guide

Technical Documentation

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Repository Initialization & Setup

git init

Initializes a new Git repository in the current directory. Creates a hidden .git folder.

```
$ git init
```

git clone

Creates a copy of a remote repository on your local machine.

```
$ git clone https://github.com/user/repo.git
```

git remote add

Connects your local repository to a remote repository URL.

```
$ git remote add origin https://github.com/user/repo.git
```

The Staging Area & Commits

git status

Shows modified, staged, and untracked files in the repository.

```
$ git status
```

git add

Stages changes before committing. Only staged files are saved in the next commit.

```
$ git add main.py
```

git commit

Records staged changes in the repository history. Always include a meaningful commit message.

```
$ git commit -m "Added login feature"
```

git diff

Shows changes between commits, branches, or working directory and staging area.

```
$ git diff
```

Branching & Merging

git branch

Lists all branches in your repository. Can also create new branches.

```
$ git branch feature-login
```

git checkout

Switches to a specific branch or commit.

```
$ git checkout feature-login
```

git merge

Merges changes from one branch into another.

```
$ git merge feature-login
```

Remote Synchronization

git push

Uploads local commits to a remote repository like GitHub.

```
$ git push origin main
```

git pull

Fetches and merges changes from the remote repository into your local repository.

```
$ git pull origin main
```

History & Undo Operations

git log

Displays the commit history of the repository.

```
$ git log
```

git reset

Undoes changes in staging area or working directory.

```
$ git reset HEAD main.py
```

git stash

Temporarily saves changes not ready to commit. Can reapply later.

```
$ git stash
```